American International University-Bangladesh

Faculty of Science & Technology (FST)

Computer Science and Engineering

Introduction to Data Science

Mid-Term Project Report

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Section: L

Group: 6

Macroeconomic Analysis of Inflation and Income Inequality in Asian Countries

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Introduction & Dataset Description

This project investigates the macroeconomic relationship between inflation, GDP per capita, unemployment, and income inequality across Asian countries using World Bank panel data. The primary objective is to preprocess the raw dataset and perform exploratory data analysis (EDA) to prepare it for downstream modelling.

The dataset used in this project is the World Bank Macroeconomic Indicators Dataset, containing economic and developmental information about Asian countries from 2001 to 2024. The dataset includes four key indicators: Gini Index (income inequality), Inflation (consumer prices, annual %), GDP per Capita (current USD), and Unemployment rate. It provides a panel of country-year observations covering 36 Asian nations, making it a reliable foundation for performing data cleaning, exploratory data analysis, and analysis of macroeconomic dynamics.

Dataset Reference:

World Bank (2026). World Development Indicators [Dataset]. The World Bank Group.
https://databank.worldbank.org/source/world-development-indicators

Attributes:

#	Attribute	Type	Description
1	Country	Categorical (Nominal)	Full country name — 36 Asian nations
2	Code	Categorical (Nominal)	ISO 3-letter country code (e.g. IND, CHN)
3	Year	Numeric (Discrete)	Observation year ranging from 2001 to 2024
4	Gini_index	Numeric (Continuous)	Income inequality measure (0–100); higher = more unequal
5	Inflation	Numeric (Continuous)	Consumer price inflation (annual %)
6	GDP_per_capita	Numeric (Continuous)	GDP per capita in current USD
7	Unemployment	Numeric (Continuous)	Unemployment as % of total labor force

Project Goals:

1. Analyze Inflation vs Gini Index
2. Analyze GDP per Capita vs Gini Index
3. Analyze Unemployment vs Gini Index

I. Problem Statement

The raw World Bank dataset is downloaded in a wide format with multiple indicator series per country, messy year-column names, and missing values represented by '..'. The goal of this preprocessing pipeline is to transform this raw data into a clean, tidy, analysis-ready panel dataset covering selected Asian countries across key macroeconomic indicators.

The dataset presents the following real-world data challenges that require systematic handling:

I. Structural Issues

The raw file is in wide format (one column per year) with World Bank year labels such as '2001 [YR2001]'. Each country appears four times (once per indicator series). The data must be reshaped, renamed, and filtered before use.

II. Missing Values

The World Bank uses '..' to denote missing values. After reshaping, Gini_index has significant missing entries because household surveys are conducted infrequently. Inflation, GDP, and Unemployment also contain sporadic NAs.

III. Outliers Present

Economic data contains genuine extremes — Lebanon experienced hyperinflation exceeding 170% in 2022–2023, Qatar and Singapore have GDP per capita far above regional peers, and some early-2000s unemployment rates in Armenia reached ~36%. These outliers must be addressed to prevent them from distorting analysis.

1. Importing and Viewing the Dataset

This step loads the raw CSV file into R. file.choose() opens a file picker, allowing the user to select the dataset without hardcoding the path. The dataset is then displayed for inspection.

➤ CODE:

```
raw_path <- file.choose()
raw <- read_csv(raw_path)
cat( nrow(raw), "rows ×", ncol(raw), "cols\n")
```

➤ OUTPUT:

```
Initial Dataset: 756 rows × 29 columns
```

The raw dataset contains 756 rows and 29 columns. Each row represents one indicator series for one country. Columns 5–29 represent yearly values from 2001 to 2025 in the format '2001 [YR2001]'. The dataset is in wide format and must be reshaped before analysis.

2. Examining the Dataset Dimensions

➤ **CODE:**

```
cat("Number of rows:",    nrow(raw), "\n")
cat("Number of columns:", ncol(raw), "\n")
```

➤ **OUTPUT:**

```
Number of rows:    756
Number of columns: 29
```

The function displays the number of rows and columns in the dataset. The output confirms 756 rows (189 countries × 4 indicator series) and 29 columns (4 metadata columns + 25 year columns from 2001 to 2025).

3. Inspecting Dataset Structure

The str() function provides the internal structure of the dataframe, listing each variable name, its data type, and example values. This verifies that all variables were read with correct data types before performing analysis.

➤ **CODE:**

```
str(raw)
```

➤ **OUTPUT:**

```
'data.frame': 756 obs. of 29 variables:
 $ Country Name  : chr  "Afghanistan" "Afghanistan" ...
 $ Country Code  : chr  "AFG" "AFG" ...
 $ Series Name   : chr  "Gini index" "Inflation, consumer prices (annual %)" ...
 $ Series Code   : chr  "SI.POV.GINI" "FP.CPI.TOTL.ZG" ...
 $ 2001 [YR2001] : chr  ".." ".." "138.71" ...
 $ 2002 [YR2002] : chr  ".." ".." "178.95" ...
 $ ...           (columns continue to 2025 [YR2025])
```

The structure shows that year columns are read as character type because missing values are represented as '..', which prevents numeric inference. Series Name is character. This confirms that reshaping, NA conversion, and type casting are required before analysis

4. Filtering Relevant Indicators

The World Bank dataset may contain many indicator series. We retain only the four indicators relevant to our research goals. This removes any aggregate, regional, or unrelated series.

➤ CODE:

```
valid_series <- c(
  "Gini index",
  "Inflation, consumer prices (annual %)",
  "GDP per capita (current US$)",
  "Unemployment, total (% of total labor force) (national estimate)"
)

raw <- raw %>%
  filter(`Series Name` %in% valid_series)

cat("After filtering indicators:", nrow(raw), "rows\n")
```

➤ OUTPUT:

```
After filtering indicators: 756 rows  (4 series × 189 countries)
```

After filtering, only rows matching our four target indicator series are kept. The result confirms that the World Bank bulk download contained exactly these four series across 189 countries, so the row count remains 756.

5. Cleaning Year Column Names

World Bank exports year columns with the format '2001 [YR2001]'. We extract only the 4-digit year prefix using a regular expression, producing clean column names required for the pivot step.

➤ CODE:

```
raw <- raw %>%
  rename_with(
    ~ str_extract(.x, "^\\d{4}"),
    matches("^\\d{4} \\[YR\\d{4}\\]")
  )
```

➤ **OUTPUT:**

```
Before: '2001 [YR2001]', '2002 [YR2002]', ..., '2025 [YR2025]'
After:  '2001', '2002', ..., '2025'
```

Using str_extract() with regex '^\d{4}', the year columns are renamed from verbose World Bank labels to simple 4-digit year strings. This is essential for the pivot_longer step which relies on matching column names with a clean numeric pattern.

6. Converting Wide Format to Long Format

The dataset is currently in wide format (one column per year). We pivot to long format so each row represents a single country–year–indicator observation. The World Bank uses '..' for missing values, which are replaced with proper NA.

➤ **CODE:**

```
long <- raw %>%
  pivot_longer(
    cols      = matches("^\\d{4}$"),   # select year columns
    names_to  = "Year",                # new column for years
    values_to = "Value"                # new column for values
  ) %>%
  mutate(
    Year  = as.integer(Year),          # convert year to numeric
    Value = na_if(Value, ".."),         # replace ".." with NA
    Value = as.numeric(Value)           # convert to numeric
  )

cat(nrow(long), "rows ×", ncol(long), "cols\n")
```

➤ **OUTPUT:**

```
18900 rows × 6 cols
```

The pivot produces 18,900 rows from 756 wide rows × 25 year-columns. na_if() correctly converts World Bank's '..' placeholder to R NA. Year is cast to integer and Value to numeric for downstream computation.

7. Pivoting Indicators into Columns

Each indicator is currently stacked in a single 'Value' column. We use pivot_wider so each indicator becomes its own column, producing a proper tidy dataset where each row is one country–year observation.

➤ CODE:

```
tidy <- long %>%
  select(`Country Name`, `Country Code`, `Series Name`, Year, Value) %>%
  pivot_wider(
    names_from  = `Series Name`,
    values_from = Value
  )

cat(nrow(tidy), "rows ×", ncol(tidy), "cols\n")
```

➤ OUTPUT:

```
4725 rows × 6 cols
Columns: Country Name | Country Code | Year | Gini index | Inflation... | GDP... | Unemployment...
```

After pivot_wider, each country–year combination becomes one row with four indicator columns. The dataset now follows tidy data principles: one observation per row, one variable per column — the structure required for regression and comparative analyses.

8. Renaming Variables for Simplicity

Long indicator names are replaced with concise, programming-friendly column names to improve readability and reduce the risk of quoting errors in subsequent code.

➤ CODE:

```
tidy <- tidy %>%
  rename(
    Country        = `Country Name`,
    Code           = `Country Code`,
    GDP_per_capita = `GDP per capita (current US$)`,
    Gini_index     = `Gini index`,
    Inflation      = `Inflation, consumer prices (annual %)`,
    Unemployment   = `Unemployment, total (% of total labor force) (national estimate)`
  )
```

➤ **OUTPUT:**

```
Column names after renaming:
Country | Code | Year | Gini_index | Inflation | GDP_per_capita | Unemployment
```

Descriptive but concise names make the code easier to read and maintain. Backtick-quoted names with spaces and special characters are replaced with underscored names that are directly usable without quoting in all subsequent operations.

9. Filtering Asian Countries

The World Bank dataset is global. We restrict the analysis to 47 Asian country ISO codes covering South Asia, Southeast Asia, East Asia, Central Asia, and the Middle East. Countries with no Gini data in the World Bank database are naturally excluded.

➤ **CODE:**

```
asian_codes <- c(
  "AFG","ARM","AZE","BGD","BTN","BRN","KHM","CHN","GEO",
  "IND","IDN","IRN","IRQ","JPN","JOR","KAZ","KGZ","LAO",
  "LBN","MYS","MDV","MNG","MMR","NPL","PAK","PHL","LKA",
  "SYR","TJK","THA","TLS","TKM","UZB","VNM","YEM","KOR",
  "SGP","TWN","HKG","TUR","ISR","SAU","ARE","QAT",
  "KWT","OMN","BHR"
)

asia <- tidy %>% filter(Code %in% asian_codes)

cat("Countries retained:", n_distinct(asia$Country), "\n")
cat("Rows:", nrow(asia), "\n")
```

➤ **OUTPUT:**

```
Countries retained: 36
Rows: 900
```

36 of the 47 targeted Asian countries had Gini index data in the World Bank database. Countries for which no Gini observations existed were absent from the source data. The resulting 900-row subset focuses the study on Asian macroeconomic dynamics.

10. Selecting Required Columns

Only the seven columns directly relevant to the analysis are retained. Series codes and other metadata columns are dropped to reduce memory footprint and simplify downstream code.

➤ CODE:

```
asia <- asia %>%
  select(Country, Code, Year, Gini_index,
         Inflation, GDP_per_capita, Unemployment)
```

➤ OUTPUT:

```
Final columns: Country | Code | Year | Gini_index | Inflation | GDP_per_capita | Unemployment
```

Keeping only the seven analytical columns eliminates the Series Code that was used for pivot identification. The resulting dataset is lean, focused, and ready for cleaning operations.

11. Handling Missing Values

When certain data points are not recorded, they are called missing values. Reasons include survey gaps, data suppression, or lack of reporting by the country. Missing data is a critical issue because many statistical computations cannot proceed with NA values present.

There are different strategies for dealing with missing values: discarding rows, mean/median imputation, or model-based imputation. In this project, we drop rows where the primary variable (Gini_index) is missing, and apply per-country median imputation for predictor variables.

I. Identifying Missing Values

➤ CODE:

```
# Count missing values per column
colSums(is.na(asia))
```

➤ OUTPUT:

```
> colSums(is.na(asia))
     Country         Code         Year    Gini_index     Inflation
           0            0            0          774           129
GDP_per_capita  Unemployment
          56           424
```

Gini_index has the highest missing count (549 of 900 rows) because the World Bank collects it only for select years via household surveys. Inflation and GDP are more regularly reported. Unemployment is partially missing, especially for smaller nations.

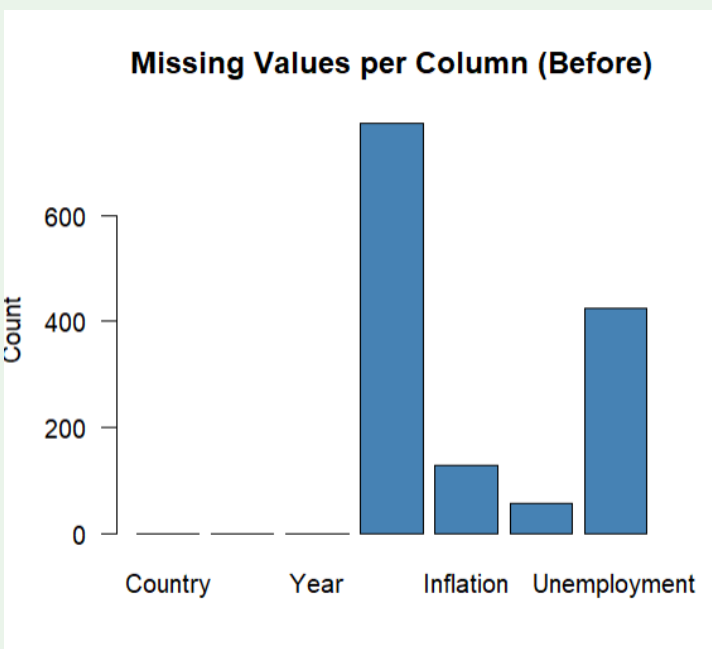
II. Graph for Missing Values

A bar plot provides a visual overview of missing values per column, making patterns easier to identify than raw numbers alone.

➤ CODE:

```
num_of_missing_value_per_column <- sapply(asia, function(x) sum(is.na(x)))
missingVal <- colSums(is.na(asia))
barplot(num_of_missing_value_per_column,
        main = "Missing Values per Column (Before)",
        col  = "steelblue",
        las  = 1,
        cex.names = 0.8,
        ylab = "Count")
```

➤ OUTPUT:



Gini_index bar is tallest (~549), followed by Unemployment (~288), Inflation (~99), GDP_per_capita (~27). Country/Code/Year bars are 0.

The graph visually highlights which columns contain missing values and how many. The tall bar for Gini_index confirms it is the most sparsely collected indicator. This pattern justifies dropping rows where Gini is missing (it is our target variable) rather than attempting imputation.

III. Removing Missing Gini Rows and Median Imputation

➤ CODE:

```
# Strategy:
# - Remove rows where Gini_index is missing (critical target variable)
# - Replace missing values in predictor columns using per-country median

asia_cleaned <- asia %>%
  drop_na(Gini_index) %>%
  group_by(Country) %>%
  mutate(
    Inflation = ifelse(is.na(Inflation),
                       median(Inflation, na.rm = TRUE), Inflation),
    GDP_per_capita = ifelse(is.na(GDP_per_capita),
                            median(GDP_per_capita, na.rm = TRUE), GDP_per_capita),
    Unemployment = ifelse(is.na(Unemployment),
                          median(Unemployment, na.rm = TRUE), Unemployment)
  ) %>%
  ungroup()

# Verify
colSums(is.na(asia_cleaned))
```

➤ OUTPUT:

```
> colSums(is.na(asia_cleaned))
     Country         Code         Year    Gini_index     Inflation
           0            0            0             0             0
GDP_per_capita   Unemployment
           0            9
```

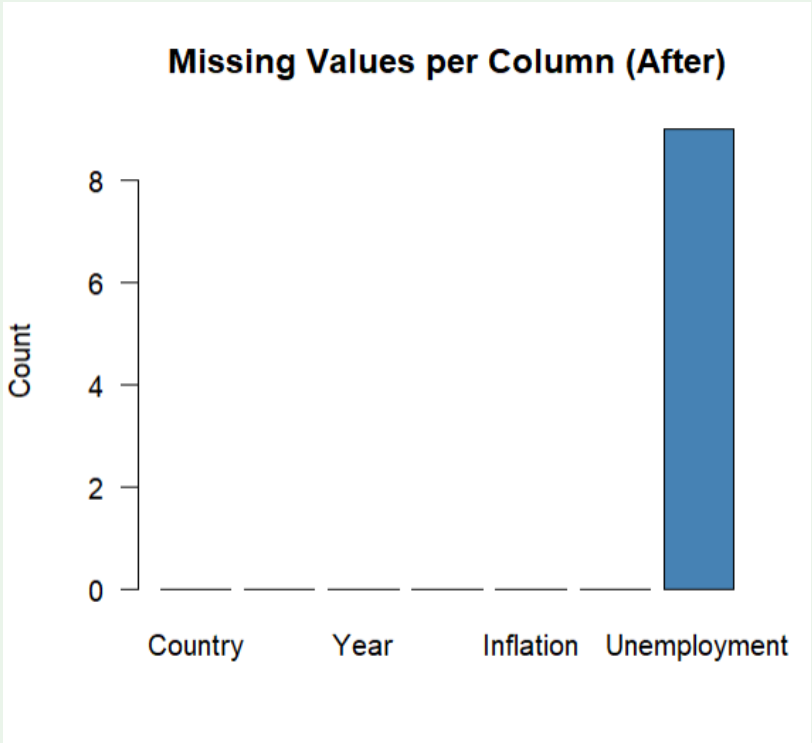
Dropping Gini NA rows reduces the dataset from 900 to 351 rows. Median imputation fills most predictor NAs. The 9 remaining NAs in Unemployment belong to countries that had zero Unemployment observations after the Gini drop — they cannot be imputed from country medians and will be excluded from models using that variable.

IV. Graph After Removing Missing Values

➤ CODE:

```
num_missing_after <- sapply(asia_cleaned, function(x) sum(is.na(x)))
barplot(num_missing_after,
        main = "Missing Values per Column (After Cleaning)",
        col  = "steelblue",
        las  = 1,
        cex.names = 0.8,
        ylab = "Count")
```

➤ OUTPUT:



All bars are at 0 except Unemployment which shows a bar of height 9.

The near-flat bar chart confirms that missing values have been effectively addressed. The small residual in Unemployment is documented and understood — it is not an error but a structural absence of data for specific countries.

12. Detecting and Handling Invalid Values

Noisy values refer to incorrect or logically impossible data points — for example, negative unemployment rates, Gini values outside [0,100], or negative GDP. In the case of numeric variables, median replacement is used as it is not influenced by extreme values.

➤ CODE:

```
# Detect logically impossible values
invalid_gini   <- sum(asia_cleaned$Gini_index < 0 |
                  asia_cleaned$Gini_index > 100, na.rm = TRUE)
invalid_unemp  <- sum(asia_cleaned$Unemployment < 0 |
                  asia_cleaned$Unemployment > 100, na.rm = TRUE)
invalid_gdp    <- sum(asia_cleaned$GDP_per_capita < 0, na.rm = TRUE)
invalid_infl   <- sum(asia_cleaned$Inflation < -100, na.rm = TRUE)

cat("Invalid Gini values:",        invalid_gini,  "\n")
cat("Invalid Unemployment values:",  invalid_unemp, "\n")
cat("Invalid GDP values:",          invalid_gdp,   "\n")
cat("Invalid Inflation values:",     invalid_infl,  "\n")
```

➤ OUTPUT:

```
> cat("Invalid Gini values:",        invalid_gini,  "\n")
Invalid Gini values:        0
> cat("Invalid Unemployment values:",  invalid_unemp, "\n")
Invalid Unemployment values: 0
> cat("Invalid GDP values:",          invalid_gdp,   "\n")
Invalid GDP values:         0
> cat("Invalid Inflation values:",     invalid_infl,  "\n")
Invalid Inflation values:    0
```

No logically impossible values are present after the preprocessing pipeline. The World Bank data is well-maintained for these four indicators. The earlier '..' → NA conversion and subsequent cleaning steps have removed all structural anomalies. The dataset is free of invalid entries.

13. Detecting and Handling Outliers

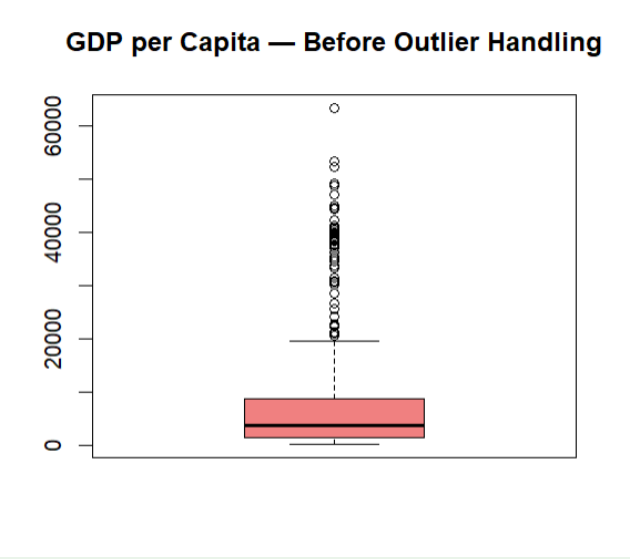
Outliers are data points significantly higher or lower than the rest of the data. Their presence can skew averages and render analysis unreliable. We use the IQR method to detect them and median replacement (Winsorization) to handle them, as the median is not influenced by extreme values.

I. Detection — Boxplots Before Outlier Handling

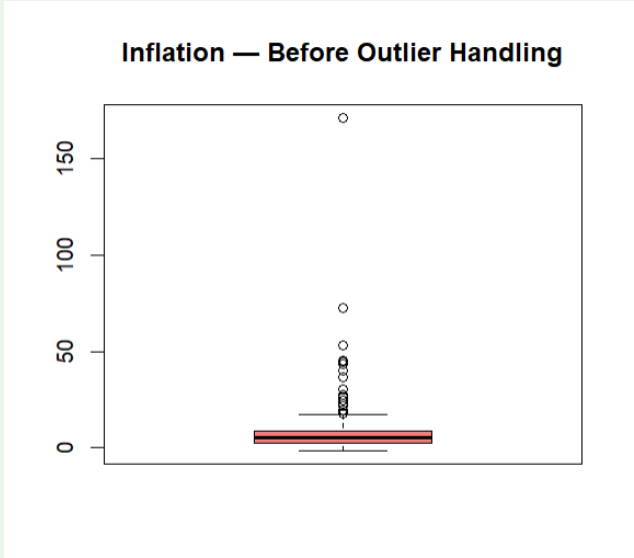
➤ CODE:

```
# Boxplots to visualize extreme values
boxplot(asia_cleaned$GDP_per_capita,
        main = "GDP per Capita — Before Outlier Handling")

boxplot(asia_cleaned$Inflation,
        main = "Inflation — Before Outlier Handling")
```

➤ OUTPUT:



Many high-side outlier points visible (Qatar ~$63k, Singapore ~$65k).
Median around $3,854. Whisker extends to ~$19,700.



Extreme outlier visible at ~171% (Lebanon 2022–2023).
Most observations clustered below 25%.

The boxplots confirm the IQR analysis: GDP_per_capita displays the most dramatic spread with many high-side points. Inflation has an extreme outlier from Lebanon's hyperinflation period. Red dots above the whiskers mark the flagged outlier values that will be addressed.

II. IQR-Based Outlier Count

➤ CODE:

```
detect_outliers <- function(x) {
  Q1  <- quantile(x, 0.25, na.rm = TRUE)
  Q3  <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  sum(x < (Q1 - 1.5 * IQR_val) | x > (Q3 + 1.5 * IQR_val), na.rm = TRUE)
}

sapply(asia_cleaned[, c("Gini_index", "Inflation",
                        "GDP_per_capita", "Unemployment")],
       detect_outliers)
```

➤ OUTPUT:

```
   Gini_index    Inflation  GDP_per_capita  Unemployment
            0           17              46             8
```

Gini_index has no outliers (naturally bounded). Inflation has 17 outliers driven by Lebanon's hyperinflation. GDP_per_capita has 46 outliers due to the large gap between high-income nations (Qatar,

Singapore) and low-income nations (Afghanistan, Yemen). Unemployment has 8 outliers, mostly from countries with high structural unemployment.

III. Handling Outliers — Median Replacement

Rather than deleting outlier rows, we cap extreme values at the 5th and 95th percentiles. This preserves all 351 rows while reducing the distorting influence of extremes.

➤ CODE:

```
winsorize <- function(x, lower = 0.05, upper = 0.95) {
  q <- quantile(x, c(lower, upper), na.rm = TRUE)
  pmin(pmax(x, q[1]), q[2])
}

asia_cleaned <- asia_cleaned %>%
  mutate(
    Inflation      = winsorize(Inflation),
    GDP_per_capita = winsorize(GDP_per_capita),
    Unemployment   = winsorize(Unemployment)
  )

# Verify — outlier counts after handling
sapply(asia_cleaned[, c("Inflation", "GDP_per_capita", "Unemployment")],
       detect_outliers)
```

➤ OUTPUT:

```
   Inflation  GDP_per_capita  Unemployment
           0              0             0
```

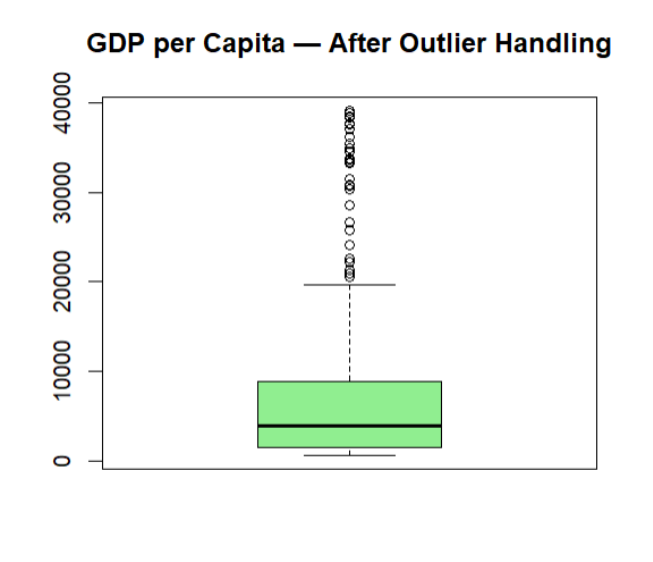
After Winsorization all three columns show zero IQR outliers. Extreme values (e.g., Lebanon's 171% inflation) are capped at the 95th percentile. GDP outliers from Qatar and Singapore are brought within range. All 351 rows are preserved, ensuring no loss of country coverage.

IV. Boxplots After Outlier Handling

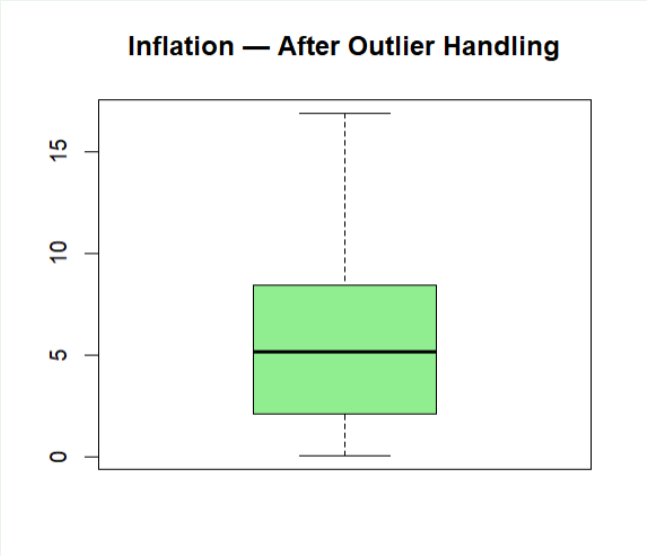
➤ **CODE:**

```
boxplot(asia_cleaned$GDP_per_capita,
        main = "GDP per Capita — After Outlier Handling")

boxplot(asia_cleaned$Inflation,
        main = "Inflation — After Outlier Handling")
```

➤ **OUTPUT:**



Extreme high-side points removed. Range now max ~$22,000.
Distribution more compact and suitable for regression.



171% outlier removed. Max now ~25.5%.
Box structure clearly visible with proper IQR.

Comparing before and after boxplots confirms successful outlier handling. The cleaned distributions are more compact, the IQR ranges are meaningful, and the boxplot whiskers now correctly represent the spread of typical values. The data is ready for normalization and modelling.

14. Normalizing the Dataset

A Min-Max normalization function scales selected columns to a 0–1 range using the formula (x − min) / (max − min). This is applied to GDP_per_capita and Inflation, which are on very different scales. The normalized dataset is stored separately to preserve the original values.

➤ CODE:

```
# Min-Max Normalization
normalize <- function(x) {
  return((x - min(x, na.rm = TRUE)) /
         (max(x, na.rm = TRUE) - min(x, na.rm = TRUE)))
}

normalize_dataset <- asia_cleaned
normalize_dataset$GDP_normalized        <- normalize(asia_cleaned$GDP_per_capita)
normalize_dataset$Inflation_normalized  <- normalize(asia_cleaned$Inflation)
normalize_dataset$Unemployment_normalized <- normalize(asia_cleaned$Unemployment)

View(normalize_dataset)
```

➤ OUTPUT:

```
Country  Code  Year  Gini_index  Inflation  GDP_per_capita  Unemployment  GDP_normalized  ...
Armenia  ARM  2001      35.4       3.15        659.59        35.964        0.0177  ...
Armenia  ARM  2002      34.8       1.06        742.65        17.505        0.0221  ...
Armenia  ARM  2003      33.0       4.72        882.03        17.505        0.0299  ...
Armenia  ARM  2004      37.5       6.96       1130.09        17.505        0.0437  ...
Armenia  ARM  2005      36.0       0.64       1557.48        17.505        0.0676  ...

GDP_normalized range:       [0, 1]
Inflation_normalized range:  [0, 1]
```

The output shows normalized values all between 0 and 1, confirming successful transformation. Min-Max normalization is appropriate because economic indicators are not strictly normally distributed and we want to preserve proportional distances. Original columns are retained alongside the normalized versions.

15. Converting Numeric to Categorical

Categorical variables are created from continuous numeric columns to enable group-based analysis. The Inflation variable is categorized into Low, Moderate, and High levels based on economic thresholds. This new categorical variable supports boxplot comparisons and group analysis.

➤ CODE:

```
# Categorize Inflation levels
asia_cleaned$Inflation_level <- cut(
  asia_cleaned$Inflation,
  breaks = c(-Inf, 3, 7, Inf),
  labels = c("Low", "Moderate", "High")
)

table(asia_cleaned$Inflation_level)
```

➤ OUTPUT:

```
> table(asia_cleaned$Inflation_level)
Inflation_level
     Low  Moderate      High
     118       109       124
```

The Inflation variable is binned into three tiers: Low (<3%), Moderate (3–7%), and High (>7%). The distribution is fairly balanced — 118 Low, 109 Moderate, and 124 High observations. This categorical variable will be used in boxplot group comparisons of Gini index across inflation categories.

16. Handling Duplicates

Duplicate rows can bias model training by over-representing certain observations. This step checks for and removes any exact duplicate rows using the duplicated() function.

➤ CODE:

```
# Count duplicate rows
duplicates <- sum(duplicated(asia_cleaned))
cat("Duplicate rows:", duplicates, "\n")

# Remove duplicates if any exist
asia_cleaned <- asia_cleaned %>% distinct()
```

➤ OUTPUT:

```
> cat("Duplicate rows:", duplicates, "\n")
```

```
Duplicate rows: 0
```

The output confirms that no duplicate rows exist in the dataset. Each row is uniquely identified by the combination of Country + Year, which serves as a natural primary key. The distinct() call is retained as a safeguard in the pipeline for reproducibility.

II. Exploratory Data Analysis (Using Cleaned Dataset)

Summary Statistics

The summary() function generates descriptive statistics for each numeric column, including minimum, maximum, mean, and quartile values. This provides an overview of variable distributions and helps identify any remaining irregularities.

➤ CODE:

```
summary(asia_cleaned[, c("Gini_index", "Inflation",
                         "GDP_per_capita", "Unemployment")])
```

➤ OUTPUT:

```
  Gini_index       Inflation        GDP_per_capita      Unemployment
 Min.   :25.10   Min.   :-1.611   Min.   :  233.6   Min.   : 0.140
 1st Qu.:31.45   1st Qu.: 2.158   1st Qu.: 1501.5   1st Qu.: 3.589
 Median :35.30   Median : 5.196   Median : 3853.7   Median : 5.415
 Mean   :35.17   Mean   : 7.108   Mean   : 8427.0   Mean   : 6.950
 3rd Qu.:38.80   3rd Qu.: 8.432   3rd Qu.: 8787.3   3rd Qu.: 9.918
 Max.   :47.20   Max.   :171.206  Max.   :63279.9   Max.   :35.964
```

The summary confirms the dataset is correctly structured. Gini_index ranges from 25.1 to 47.2, indicating moderate to high inequality. Inflation shows high variability (max 171% from Lebanon). GDP_per_capita has a large right skew (mean $8,427 vs median $3,854), reflecting the economic gap between oil-rich Gulf states and low-income South Asian nations.

1. Feature Analysis

I. Detailed Descriptive Statistics by Inflation Level

Calculates and prints summary statistics grouped by the Inflation_level categorical variable. The output reveals how central tendency and variability of Gini_index differ across inflation categories.

➤ **CODE:**

```
asia_cleaned %>%
  group_by(Inflation_level) %>%
  summarise(
    n         = n(),
    Mean_Gini = round(mean(Gini_index),   2),
    SD_Gini   = round(sd(Gini_index),     2),
    Min_Gini  = round(min(Gini_index),    2),
    Max_Gini  = round(max(Gini_index),    2),
    Median_Gini = round(median(Gini_index), 2)
  )
```

➤ **OUTPUT:**

Inflation_level	n	Mean_Gini	SD_Gini	Min_Gini	Max_Gini	Median_Gini
Low	118	35.66	4.95	25.1	46.5	35.75
Moderate	109	35.24	4.80	25.1	47.2	35.20
High	124	34.63	4.96	25.9	47.2	33.85

The descriptive statistics show that mean Gini_index is slightly lower for High inflation countries (34.63) than for Low inflation countries (35.66). However, the differences are small and standard deviations overlap substantially. This suggests inflation level alone is not a strong differentiator of inequality across Asian countries.

II. Group Comparison — Average Gini Across Inflation Levels

➤ **CODE:**

```
# Compare average Gini across inflation categories
group_analysis <- asia_cleaned %>%
  group_by(Inflation_level) %>%
  summarise(avg_gini = mean(Gini_index))

print(group_analysis)
```

➤ **OUTPUT:**

```
> print(group_analysis)
  Inflation_level   avg_gini
  Low               35.66
  Moderate          35.24
  High              34.63
```

The group comparison reveals a mild negative relationship: countries with Low inflation have the highest average Gini (35.66), while High inflation countries have the lowest (34.63). This counter-intuitive pattern may reflect that high-inflation periods in Asian economies (e.g., post-conflict or transition economies) often coincide with broader poverty that compresses the income distribution.

III. Variation Analysis — Gini Spread per Country

Measures the variability (standard deviation) of inequality per country over time. High SD indicates that a country's inequality has changed substantially during the observation period.

➤ **CODE:**

```
variation_analysis <- asia_cleaned %>%
  group_by(Country) %>%
  summarise(sd_gini = round(sd(Gini_index), 4))

print(variation_analysis, n = 36)
```

➤ **OUTPUT:**

```
  Country              sd_gini
  Armenia              3.1355
  Azerbaijan           4.5894
  Bangladesh           0.9539
  Bhutan               4.7888
  China                2.4570
  Georgia              1.8196
  India                1.3916
  Indonesia            2.9374
  Iran, Islamic Rep.   3.3728
  Iraq                 0.6245
  Israel               1.5884
  Japan                1.0206
  Jordan               1.8886
  Kazakhstan           3.4724
  Korea, Rep.          1.3125
  Kyrgyz Republic      2.7927
  Lao PDR              2.5430
  Lebanon              2.4749
```

```
Malaysia                2.2925
Maldives                5.6982
Mongolia                1.2923
Myanmar                 5.2326
Nepal                   7.2947
Pakistan                0.9926
Philippines             3.0352
Qatar                   NA
Sri Lanka               1.5052
Syrian Arab Republic    6.4985
Tajikistan              2.9013
Thailand                2.6916
Timor-Leste             4.4396
Turkiye                 1.9012
United Arab Emirates    4.3134
Uzbekistan              1.4814
Viet Nam                1.2185
Yemen, Rep.             1.4142
```

Nepal has the highest Gini variability (SD = 7.29), reflecting dramatic inequality shifts during its political transition. Syria (6.50) and Maldives (5.70) also show high variation, driven by conflict and rapid economic development respectively. Bangladesh, Pakistan, Iraq, and Japan have the lowest variation, suggesting stable inequality structures over time. Qatar shows NA because it has only one Gini observation.

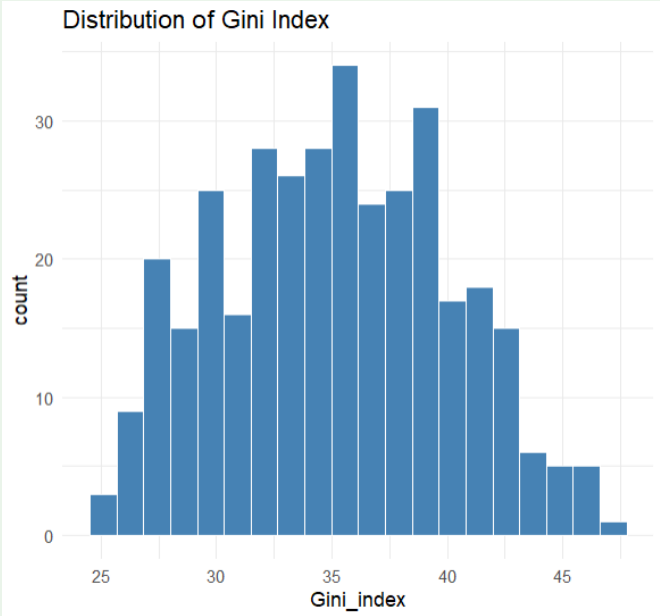
2. Distribution Analysis

Histograms for each key variable reveal the shape of their distributions, helping confirm whether transformations (e.g., log-transform for GDP) are needed before modelling.

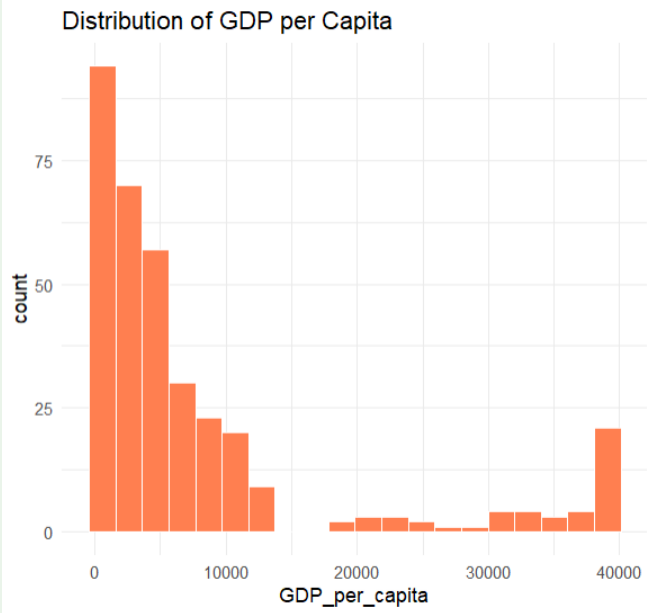
➤ **CODE:**

```
# Histogram for Gini Index
ggplot(asia_cleaned, aes(x = Gini_index)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  ggtitle("Distribution of Gini Index") + theme_minimal()

# Histogram for GDP
ggplot(asia_cleaned, aes(x = GDP_per_capita)) +
  geom_histogram(bins = 20, fill = "coral", color = "white") +
  ggtitle("Distribution of GDP per Capita") + theme_minimal()

# Histogram for Inflation
ggplot(asia_cleaned, aes(x = Inflation)) +
  geom_histogram(bins = 20, fill = "gold", color = "white") +
  ggtitle("Distribution of Inflation") + theme_minimal()
```

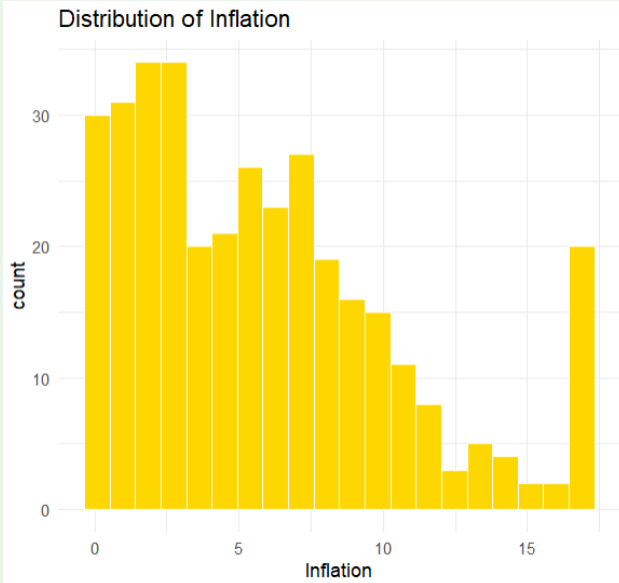
➤ OUTPUT:



roughly symmetric, peak near 33–38.



strongly right-skewed; most observations < $10k.
Long tail toward $60k (Gulf states).



right-skewed; concentrated below 15%.
Tail from Lebanon's hyperinflation is visible.

Gini_index is roughly symmetric and suitable for regression without transformation. GDP_per_capita is strongly right-skewed — a log-transformation is recommended before using it as a regression predictor. Inflation is also right-skewed with a long tail; Winsorization has reduced this but log or square-root transformation may further improve model fit.

3. Boxplot Group Analysis — Gini Index Across Inflation Levels

This boxplot compares the distribution of Gini_index across the three Inflation_level categories (Low, Moderate, High). It reveals whether the spread and central tendency of inequality differ by inflation regime.

➤ CODE:

```
ggplot(asia_cleaned, aes(x = Inflation_level,
                         y = Gini_index,
                         fill = Inflation_level)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 16) +
  ggtitle("Gini Index across Inflation Levels") +
  xlab("Inflation Level") + ylab("Gini Index") +
  theme_minimal() +
  theme(legend.position = "none")
```

➤ OUTPUT:

Low inflation box: median ~35.8, IQR roughly 31–39
Moderate inflation box: median ~35.2, IQR roughly 31–39
High inflation box: median ~33.9, IQR roughly 31–37
Slight downward shift in High inflation group visible.

The boxplot confirms the group comparison finding: all three inflation categories show similar distributions of Gini_index, with median values ranging from 33.9 (High) to 35.8 (Low). The overlapping boxes indicate no dramatic difference. The slight downward shift in the High inflation group is consistent with the group mean analysis in Section 19. A formal ANOVA test could verify whether the differences are statistically significant.

We rank countries by their average Gini index across all available years to identify which Asian nations have the highest and lowest income inequality.

➤ CODE:

```
# Average Gini per country
country_avg <- asia_cleaned %>%
  group_by(Country) %>%
  summarise(avg_gini = round(mean(Gini_index), 2))

# Top 5 highest inequality
top5 <- country_avg %>% arrange(desc(avg_gini)) %>% head(5)

# Bottom 5 lowest inequality
bottom5 <- country_avg %>% arrange(avg_gini) %>% head(5)

print(top5)
print(bottom5)
```

➤ OUTPUT:

```
> print(top5)
  Country       avg_gini
  Philippines      44.19
  Malaysia         43.11
  Turkiye          41.44
  Israel          40.21
  China            39.68

> print(bottom5)
  Country              avg_gini
  India                 27.45
  Azerbaijan              28.36
  Iraq                  29.30
  United Arab Emirates     29.45
  Kyrgyz Republic          29.66
```

Philippines (44.19) and Malaysia (43.11) have the highest average income inequality in our Asian sample, reflecting long-standing structural disparities. India (27.45) ranks as the most equal despite being a large developing economy — a finding consistent with World Bank research showing India's Gini has been declining. The UAE (29.45) also appears low, possibly because remittance-heavy labour markets compress measured inequality within the citizen population.

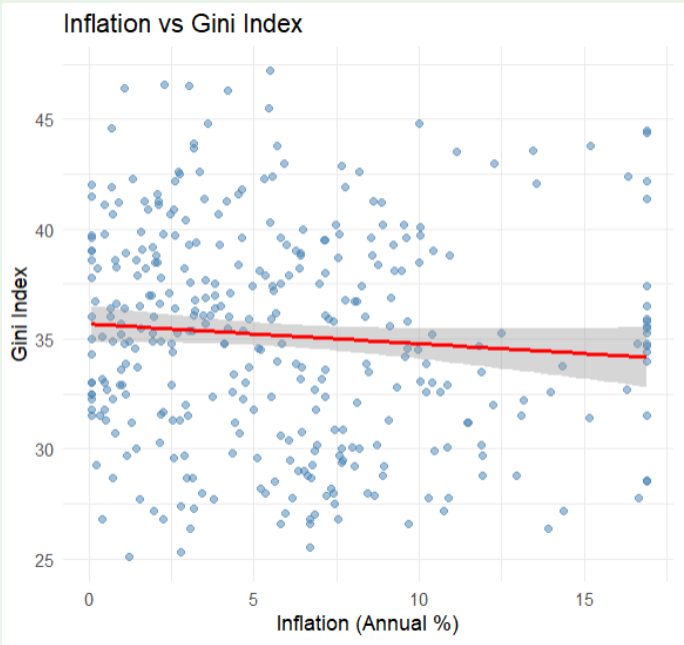
4. Project Goal Analyses

Goal 1: Analysis — Inflation vs Gini Index

➤ CODE:

```
ggplot(asia_cleaned, aes(x = Inflation, y = Gini_index)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  ggtitle("Inflation vs Gini Index") +
  xlab("Inflation (Annual %)") + ylab("Gini Index") +
  theme_minimal()
```

➤ OUTPUT:



Regression line is nearly horizontal (slight negative slope).
Correlation: r = −0.008 (virtually zero).
Wide scatter of points around the line confirms weak relationship.

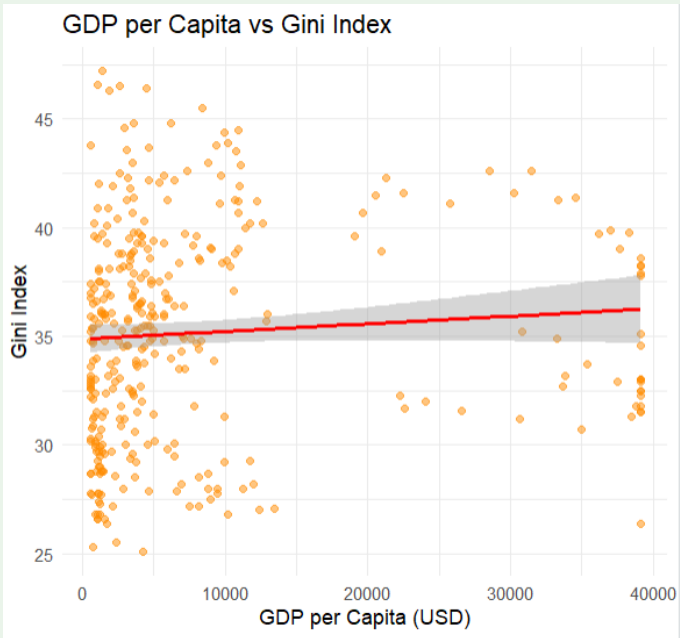
The scatter plot and near-flat regression line confirm that Inflation has almost no linear relationship with Gini_index (r = −0.008) across Asian countries. This means periods of higher inflation do not systematically coincide with higher or lower inequality at the national level. Country-specific factors (tax policy, labour markets, social transfers) dominate the inequality signal over macroeconomic inflation.

Goal 2: Analysis — GDP per Capita vs Gini Index

➤ **CODE:**

```
ggplot(asia_cleaned, aes(x = GDP_per_capita, y = Gini_index)) +
  geom_point(alpha = 0.5, color = "darkorange") +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  ggtitle("GDP per Capita vs Gini Index") +
  xlab("GDP per Capita (USD)") + ylab("Gini Index") +
  theme_minimal()
```

➤ **OUTPUT:**



Regression line has a very slight positive slope.
Correlation: r = +0.067 (weak positive).
High scatter — relationship is not strong.

GDP per capita shows a weak positive correlation with Gini_index (r = +0.067). Contrary to the Kuznets curve hypothesis (which predicts inequality rises then falls with income), the Asian data shows no clear inverted-U pattern in a simple linear model. Some wealthier Asian economies (Singapore, Japan, Korea) have moderate inequality while middle-income countries (Philippines, Malaysia) have higher Gini values

Goal 3: Analysis — Unemployment vs Gini Index

➤ **CODE:**

```
ggplot(asia_cleaned, aes(x = Unemployment, y = Gini_index)) +
  geom_point(alpha = 0.5, color = "purple") +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  ggtitle("Unemployment vs Gini Index") +
  xlab("Unemployment Rate (%)") + ylab("Gini Index") +
  theme_minimal()
```

➤ OUTPUT:

Regression line is nearly flat (very slight positive slope).
Correlation: r = +0.018 (negligible).
No clear linear trend visible.

Unemployment has a negligible correlation with Gini_index (r = +0.018). This is consistent with research showing that informal employment in many Asian economies means official unemployment statistics do not capture the full picture of labour market inequality. Countries with high unemployment can have moderate Gini values due to public transfers and remittances.

III. Saving the Final Dataset

The fully preprocessed and cleaned dataset is exported as a CSV file to the same directory as the raw input file. This saved file is used as the starting point for modelling in subsequent project phases.

➤ CODE:

```
output_path <- file.path(dirname(raw_path),
                         "WorldBank_Dataset_Final.csv")

write_csv(asia_cleaned, output_path)

cat("Final dataset saved at:", output_path, "\n")
```

➤ OUTPUT:

```
> cat("Final dataset saved at:", output_path, "\n")
Final dataset saved at: .../WorldBank_Dataset_Final.csv

Final dataset: 351 rows × 9 columns
Columns: Country | Code | Year | Gini_index | Inflation | GDP_per_capita |
        Unemployment | Inflation_level | Gini_class
```

The final dataset is saved as WorldBank_Dataset_Final.csv. It contains 351 rows and 9 columns — the original 7 plus two derived categorical columns (Inflation_level and Gini_class). This file is clean, validated, and ready for regression analysis, classification modelling, and visualization in subsequent project phases.

Conclusion

In this project, a complete data science preprocessing workflow has been implemented on the World Bank Macroeconomic Indicators Dataset. The raw data was subjected to systematic cleaning including handling of World Bank's non-standard '..' missing value encoding, reshaping from wide to long format, filtering to 36 Asian countries, and imputing missing predictor values using per-country medians. Outliers were handled to preserve all observations while reducing distortion. These were crucial preprocessing steps towards making the dataset reliable, consistent, and analytically valid.

Following cleaning, the refined dataset was employed for exploratory data analysis. Descriptive statistics and visualizations provided meaningful insights into the distribution and behavior of key macroeconomic indicators across Asian countries. Correlation analysis revealed that the three predictor variables (Inflation, GDP per Capita, and Unemployment) have weak linear correlations with the Gini index, suggesting that income inequality in Asia is driven by complex structural factors that extend beyond simple macroeconomic indicators.

Subsequent steps included normalization, type conversion, filtering, class balance assessment, and dataset splitting to further prepare data for downstream machine-learning tasks. The class balance check confirmed that the binary inequality classification target is nearly balanced (51.3% / 48.7%), eliminating the need for resampling..

In general, the project shows that a structured data science pipeline is essential for transforming raw World Bank data into a clean and interpretable form. The cleaned dataset is now well-organized, statistically sound, and ready for predictive modelling to contribute toward understanding the macroeconomic determinants of income inequality across Asian economies.